

Tanay Ketan Lavingia

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Education

Master of Science in Mechanical Engineering <i>University of Massachusetts Lowell</i> MA	December 2026
Relevant Courses: Plastic Product Design, Rapid Prototyping, Applied Finite Element Analysis, Solid Mechanics	
Bachelor of Engineering in Mechanical Engineering <i>Gujarat Technological University</i> Ahmedabad, India	June 2023

Skills

CAD/CAE: SolidWorks, AutoCAD, Ansys (FEA, CFD), MATLAB, ABAQUS, Altair Inspire, Moldflow
Manufacturing: CNC Milling, 3D Printing, Laser Cutting, Power tools
Productivity: Technical Documentation, MS Office, Minitab

Work Experience

Fresenius Medical Care North America | Lawrence, MA

Mechanical Engineering Co-Op

July 2025 – December 2025

- Designed test fixtures in SolidWorks, integrating fluid pathways and valve assemblies, fabricated prototypes using 3D printing and CNC milling to support testing of next-generation electromechanical peritoneal dialysis cyclers components
- Built and validated thermofluid simulation in ANSYS matching current component thermal behavior, creating a reusable baseline for evaluating future design variants
- Performed static structural simulations in ANSYS to evaluate stress distribution and structural integrity, guiding design modifications to improve component performance
- Characterized plastic film material properties using Instron Tensile Tester to generate validated FEA inputs, enabling predictive structural analysis of medical device components
- Collaborated with human factors team and hand-modified prototype cyclers door components to test ergonomic improvements, incorporating validated changes into the next design revision

3Brothers Machination | Ahmedabad, India

Project Coordinator - Machine Design and Development

July 2023 – July 2024

- Coordinated a cross-functional team to redesign a hydraulic cold forging machine, reworking the hydraulic cylinder and loading points, increasing reliability by 30% and improving logistical safety
- Designed a 4WD transfer case for an off-road prototype vehicle, coordinated component fabrication and machining, and assembled the final unit
- Developed 3D models, detailed technical drawings, and BOMs in SolidWorks, applying GD&T and optimizing designs for manufacturability and assembly (DFM/DFA)
- Resolved a recurring failure by identifying incorrect GD&T specifications as the root cause and implementing a CAPA, extending component life from 5–6 shots to the full rated production duration
- Supported machine assembly and shop-floor activities by inspecting manufactured parts, collaborated with machinists, and implemented design revisions based on manufacturing feedback.

Academic Projects

Design and Prototyping of a dual lemon juicer (Course: Rapid Prototyping)

Spring 2026

- Designed a 7-component lime juicer assembly in CAD, incorporating integrated strainers, guided compression mechanisms, and ergonomic features to support household and small-scale commercial use
- Fabricated and assembled functional prototypes using FDM 3D printing, performing tolerance analysis and fit validation for mating components and sliding shaft mechanisms
- Evaluated prototype performance through functional testing and root-cause analysis, implementing design modifications including structural ribs, increased support features, and geometry changes to improve durability and manufacturability

Case study of Water Filtration System (Course: Plastic Product Design)

Spring 2026

- Created CAD models of the complete water filtration assembly, including the basin, reservoir, lid, and associated components, to evaluate product design, assembly interfaces, and injection molding requirements
- Analyzed materials by balancing impact resistance, thermal performance, food-safety requirements, cost, and manufacturability for appropriate material selection
- Performed Moldflow simulations on lid and reservoir components to analyze fill behavior, shrinkage, cooling performance, pressure distribution, and molding defects such as weld lines and warpage.